COLLINSON

PriModel

Recommendation Engine
Waitlist LMS

Document Structure

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Problem & Framework

Part I

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The Four Pillars of Prioritization

PriModel Engine

Capacity

Minimizing seat-minutes and maximizing throughput per available square foot.

Fairness

Protecting the passenger experience through accumulated wait-time tracking.

Urgency

Smart flight mapping to optimize usable lounge time.

Revenue

Maximizing commercial yield per admission based on program value.

System Suggests, Staff Decides.

1.1 The Question PriModel Answers

Given N active waitlist entries, each with different commercial value, party size, flight departure time, and time spent waiting: **which entry should the staff call next?**

Advisory-only recommendation engine for lounge staff

Advisory only

PriModel suggests, staff decides

Lounge-scoped

Every parameter configurable per lounge

Graceful degradation

Missing data = neutral (50), never penalties

1.2 Why FIFO Is Insufficient

Capacity Blindness

Solo (20 min stay) vs group of 3 (2.5h stay). With 3 seats, calling the group locks all capacity. 1 admission vs 4.

Revenue Blindness

A $175 group vs a $24 group. Over a peak shift, thousands in incremental revenue. FIFO leaves it to chance.

Fairness Blindness

Passenger waiting 40 min, skipped 8 times vs one waiting 5 min. FIFO only sees arrival order.

Fit Blindness

95% capacity, 2 seats, group of 4 first in line. They do not fit. Lounge freezes. 4 solos could be served.

FIFO vs PriModel: The Capacity Impact

3 seats available. Group A: solo, flight in 50 min. Group C: party of 3, flight in 3h.

Path 1: FIFO

Calls Group C (party of 3) based on arrival order

All 3 seats locked for 2.5 hours

450 seat-minutes consumed
Lounge frozen. Group A misses flight. 1 admission.

Path 2: PriModel

Calls Group A (solo) based on capacity impact

Solo uses 1 seat for 20 min, then frees it

20 seat-minutes consumed (22x less)
Seat reopens at 3:40 PM. 2+ admissions.

1.3 Design Constraints

Advisory only	PriModel suggests, staff decides. No automated admission.
Lounge-scoped	Every parameter is configurable per lounge.
Graceful degradation	Missing data produces neutral scores (50), never penalties.
Fairness floor	No configuration can reduce fairness weight below 0.10.
Passenger-invisible	Passengers never see scores, dimensions, or prioritization logic.
Full parametrization	Every lookup table value, boundary, and weight is admin-configurable without code changes.

The Four Pillars of Prioritization

PriModel Engine

Capacity

Minimizing seat-minutes and maximizing throughput per available seat.

Fairness

Protecting passenger experience through accumulated wait-time tracking.

Urgency

Smart flight mapping to optimize usable lounge time. No taper.

Revenue

Maximizing commercial yield per admission based on program value.

System Suggests, Staff Decides.

System Ontology: Inputs & Variables

Category	Variables
Real-Time Environment State	**available_seats – derivated (flight data)** pre_bookings_N_mins
Dynamic Entity State	wait_time (elapsed mins) flight_status **time_to_departure (mins)- derivated** access_program
Derived Entity State	estimated_stay / ETA day_of_week pattern (derived ETA) **value_per_person** (benefit card / membership ID) [total_group_value (derived from value/pp)

Visit Fee Reference	
Day Pass (walk-in) *No issuer deal – lounge retains full margin*	**$35**
Priority Pass *Standard issuer deal rate*	**$27**
Amex Platinum *Issuer-negotiated rate*	**$25**
LoungeKey *Standard issuer deal rate*	**$24**

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2. Scoring Dimensions

PriModel evaluates each active waitlist entry across four dimensions. Each produces a sub-score normalized to [0, 100] via parametrized lookup tables.

Revenue

How much is this group worth?

Core metric:

Value/pp lookup (0-100)

Example gap:

Day Pass $35 = 100 vs LK $24 = 60

Capacity

How much space/time consumed?

Core metric:

Party + Stay lookup (0-100)

Example gap:

Solo 20min = 100 vs Group 3h = 37

Fairness

How long waiting? How many skips?

Core metric:

Wait + Skips lookup (0-100)

Example gap:

40 min = 80 vs 5 min = 10

Urgency

How close is the flight?

Core metric:

TTD lookup (0-100). No taper.

Example gap:

TTD 15 min = 100 vs 180+ = 0

3. Variables (Input Data)

#	Variable	Dimension	Source	Notes
1	Occupancy %	Capacity	LMS headcount vs max	Fallback: staff toggle (L/M/H)
2	Wait time (min)	Fairness	current_time - join_ts	Always available
3	Party size	Revenue, Capacity	Enrollment field	Always available
4	Pre-bookings (next N min)	Capacity	Pre-booking system	Fallback: 0
5	Flight status	Urgency	Flight API real-time	On Time / Delayed / Boarding / Departed
6	Time to departure (min)	Urgency, Capacity	Derived from flight	Flight number mandatory
7	Access program	Revenue	Passenger/staff declared	Frequently missing
8	Estimated stay (min)	Capacity	Derived from #6	Fallback: 120 min default
9	Day-of-week pattern	Capacity (context)	Historical analytics	Phase 3+
10	Value per person	Revenue	Benefit card / deal lookup	Hard to get
11	Total group value	Revenue	value/pp x party_size	Derived from #10

Flight number is mandatory in enrollment. Variables #5, #6, #8 are almost always available. Primary data gaps: #7 (access program) and #10/#11 (value/pp).

Operational Scenarios

Part II

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4. Scenarios Overview

5 scenarios demonstrate PriModel behavior in real lounge situations. Core principle: the system suggests, staff decides.

1 **Capacity Efficiency Beats Raw Revenue**

More admissions/hour

2 **Fairness Protects Long-Waiting Passengers**

Fewer complaints, better brand

3 **Flight Urgency and Short Visits Are Optimal**

Short visit = fee + fast turnover

4 **Revenue Optimization When Capacity Allows**

Highest-value group captured. $175 in one transaction.

5 **Peak Hour Throughput Machine**

4 admissions in 45 min vs 1

Test Case 1: Capacity Efficiency vs. Raw Revenue

GLOBAL STATE

3:20 PM | Available: 3 Seats | Mode: Balanced

Group A – Priority Pass

Party: 1 | $27/pp | Total: $27

Flight: 4:10 PM (50 min) *Wait: 15 min*

Group B – Amex Platinum

Party: 2 | $25/pp | Total: $50

Flight: 5:20 PM (2h) *Wait: 10 min*

Group C – LoungeKey

Party: 3 | $24/pp | Total: $72

Flight: 6:20 PM (3h) *Wait: 5 min*

ANALYSIS

Why Capacity Wins

Seat-minutes comparison:
Group A: 20 seat-minutes
Group B: 180 seat-minutes
Group C: 450 seat-minutes (22x Group A)

Revenue gap A vs C: $45
A $45 advantage cannot overcome a 22x capacity penalty.

Calling A: seat opens at 3:40 PM.
Calling C: all 3 seats locked until 6 PM.

OUTPUT RANK

#1

Group A – Priority Pass solo

"Quick turnover + flight departing soon"

#2

Group B – Amex Platinum couple

"Balanced value and capacity"

#3

Group C – LoungeKey party of 3

"Highest commercial value"

Without PriModel: a group of 3 arriving 2 min earlier freezes the lounge until 6 PM. With PriModel: the solo creates immediate turnover — 2 admissions instead of 1.

Test Case 2: The Fairness Multiplier

GLOBAL STATE

Available: 4 Seats | Mode: Balanced

Group A – Priority Pass couple

Party: 2 | $27/pp | Total: $54

Flight: 5:15 PM (1h) *Wait: 35 min*

Group B – Amex Platinum (x4)

Party: 4 | $25/pp | Total: $100

Flight: 6:15 PM (2h) *Wait: 5 min*

Group C – LoungeKey solo

Party: 1 | $24/pp | Total: $24

Flight: 7:15 PM (3h) *Wait: 40 min*

ANALYSIS

Why Fairness Dominates

Fairness gap: 8x
Groups A & C waiting 35-40 min vs Group B at 5 min.

A gets #1 over C: combines strong fairness with urgency (flight in 1h).

Group B's $100 is the highest revenue, but in Balanced mode fairness outweighs revenue.

B will climb as their wait increases — but **cannot skip passengers waiting 8x longer.**

OUTPUT RANK

#1

Group A – Priority Pass couple

"Long wait + flight departing soon"

#2

Group C – LoungeKey solo

"Longest waiting group"

#3

Group B – Amex Platinum (x4)

"Highest commercial value"

PriModel creates equilibrium: premium programs cannot blindly jump the queue. The $100 group is #3 and climbs as their wait increases — but never skips passengers waiting 8x longer.

Test Case 3: Smart Urgency & The Usable Time Curve

GLOBAL STATE

Available: 2 Seats | Mode: Balanced

Group A – LoungeKey couple

Party: 2 | $24/pp | Total: $48

Flight: 5:45 PM (1h 30min) *Wait: 20 min*

Group B – Priority Pass couple

Party: 2 | $27/pp | Total: $54

Flight: 7:15 PM (3h) *Wait: 15 min*

Group C – Amex Platinum solo

Party: 1 | $25/pp | Total: $25

Flight: 4:55 PM (40 min) *Wait: 10 min*

ANALYSIS

The Urgency Taper Logic

Key question:
Should 40 min = 90 min to departure?

Answer: No. Urgency scales with usable time.

Group A (90 min to departure):
Optimal usable window (~60 min visit). Best combo: longest wait + urgency within the window.

Group C (40 min to departure):
Nearest flight ≠ highest priority. Receives a tapered urgency boost (~89%) because a 10-min visit generates less operational value than 60 min.

Staff can override and call C if gate is nearby. No friction. The system ranks; the staff judges.

OUTPUT RANK

#1

Group A – LoungeKey couple

"Longest wait + flight within optimal window"

#2

Group C – Amex Platinum solo

"Low seat impact + near departure"

#3

Group B – Priority Pass couple

"Moderate commercial value"

PriModel does not blindly chase the nearest departure. Urgency scales with usable lounge time. Flight info is always visible to staff — the system ranks, the staff judges.

Test Case 4: Revenue Optimization When Capacity Allows

GLOBAL STATE

Available: 5 Seats | Revenue Focus Active
AD-operated lounge – commercial yield is a KPI

Group A – LoungeKey solo
Party: 1 | $24/pp | Total: $24
Flight: 6:15 PM (2h) *Wait: 20 min*

Group C – Priority Pass couple
Party: 2 | $27/pp | Total: $54
Flight: 6:15 PM (2h) *Wait: 18 min*

Group B – Day Pass party of 5
Party: 5 | $35/pp | Total: $175
Flight: 7:15 PM (2h) *Wait: 15 min*

ANALYSIS

Why Revenue Dominates Here

Group B: $175 (7x Group A, 3x Group C)

In Revenue Focus, revenue carries ~50% of the total score formula.

This works because capacity allows it:
5 seats available, Group B needs exactly 5. Capacity constraint is mathematically nullified.

Safeguard: if any wait crosses 60 min, hard override fires and that group becomes #1.

OUTPUT RANK

#1
Group B – Day Pass party of 5
"Highest commercial value ($175)"

#2
Group C – Priority Pass couple
"Balanced value and capacity ($54)"

#3
Group A – LoungeKey solo
"Longest waiting group ($24)"

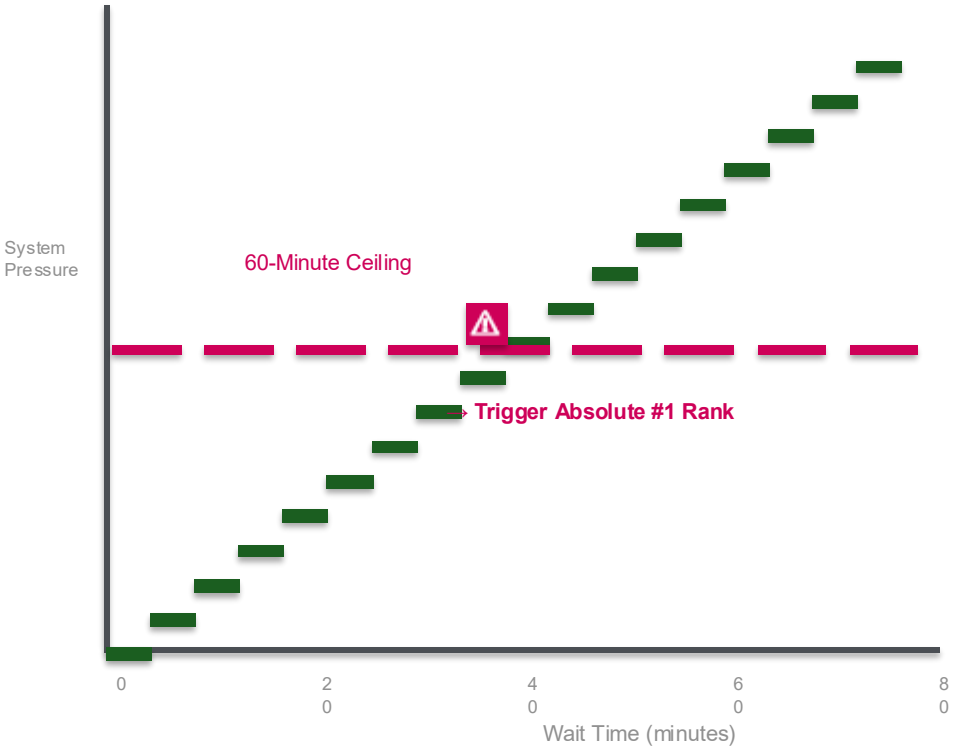
$175 captured in a single transaction.

Revenue Focus moves the highest-value group to #1 when capacity allows. Group A is still visible in top 3. If their wait crosses 60 min, the hard override fires.

System Focus Modes: Weight Manipulation Matrix

	Efficiency Focus	Balanced Focus	Revenue Focus
Capacity (Cobalt)	Heavily dominates. Max throughput, min seats.	Optimizes flow.	Constraint nullified if seats > party size.
Fairness (Forest Green)	Acts only as a tie-breaker.	Anchor. Long waits always win ties.	Standard scaling.
Urgency (Amber)	Crucial for turnover.	Optimizes flow.	De-prioritized.
Revenue (Crimson)	Nearly zeroed out.	Tie-breaker, not a driver.	Grows exponentially, up to ~50% of formula.

The Hard Override Mechanism



The Vulnerability

In Efficiency or Revenue modes, high-capacity or low-yield groups can mathematically stall against optimal profiles indefinitely.

The Failsafe

max_wait_ceiling = 60_minutes.
When wait_time >= 60, the algorithm triggers an absolute #1 rank override, bypassing all other logic weights.

Operational Context

By the time the override fires for stalled groups (~5:00 PM at GRU), the departure wave has passed, seats naturally open, and they are seated without destroying peak throughput.

Test Case 5: The Peak Throughput Machine – GRU Departure Wave

Location: GRU – Friday, 4:15 PM
Flights: MIA, EZE, SCL, LIM, BOG (5:00–6:30 PM)
Status: 95% Capacity | 18 Groups Waiting
Mode: Efficiency Focus Active

Top 6 in Queue (FIFO order):

Grp	Program	Size	$/pp	Flight	Wait
C	Day Pass	2	$35	AA920 MIA (42m)	42min
A	Priority Pass	1	$27	LA635 SCL (45m)	38min
E	Priority Pass	1	$27	AR1161 EZE (55m)	30min
B	Airline Biz	1	$27	AV109 BOG (50m)	22min
F	Airline Biz	1	$27	LA2480 LIM (1h)	18min
D	LoungeKey	4	$24	AA920 MIA (3h)	15min

PriModel Output – Efficiency Focus

#1 Priority Pass solo – LA 635 to SCL

"Quick turnover + flight departing soon"

#2 Priority Pass solo – AR 1161 to EZE

"Quick turnover + flight departing soon"

#3 Airline Business solo – AV 109 to BOG

"Quick turnover + flight departing soon"

The Logic:

Solos: 15–25 seat-minutes each.
Group C: 300 seat-min | Group D: 360 (doesn't fit 2 seats).
Capacity gap: 12–20x. Revenue identical ($27) and de-weighted.
A > E: waited longer (38 vs 30 min) — fairness breaks the tie.

4 admissions through 2 seats in 45 min. $108 vs $70 from a single long-stay admission.

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Algorithmic Synthesis: The Objective Function

$$\text{Priority_Score} = (w1 \times F(t)) + (w2 \times U(d)) + (w3 \times C(s,o)) + (w4 \times R(v))$$

w1–w4: Dynamic weights controlled by the active Focus Mode state

F(t)

Fairness

Geometric scaling based on elapsed wait time. Long waits grow exponentially to prevent starvation.

U(d)

Urgency

Tapered usable-time curve based on departure proximity. Peaks at ~90 min, tapers on both extremes.

C(s,o)

Capacity

Inverse penalty based on party size vs available seats. High-capacity groups penalized when seats are scarce.

R(v)

Revenue

Linear score based on program value per person. Exponentially scaled in Revenue Focus mode.

↓

Output: A sorted array presented to lounge staff as a simple, actionable Top 3 list.

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Test Case 1: Capacity Efficiency vs. Raw Revenue

GLOBAL STATE

3:20 PM | Available: 3 Seats | Mode: Balanced

Group A - Priority Pass
Party: 1 | $27/pp | Total: $27
Flight: 4:10 PM (50 min) | Wait: 15 min

Group B - Amex Platinum
Party: 2 | $25/pp | Total: $50
Flight: 5:20 PM (2h) | Wait: 10 min

Group C - LoungeKey
Party: 3 | $24/pp | Total: $72
Flight: 6:20 PM (3h) | Wait: 5 min

ANALYSIS

Why Capacity Wins

Seat-minutes comparison:
Group A: 20 seat-minutes (Capacity: 100)
Group B: 180 seat-minutes (Capacity: 67)
Group C: 450 seat-minutes (Capacity: 37)

A $45 revenue gap cannot overcome a 63-point capacity gap.

Calling A: seat opens at 3:40 PM.
Calling C: all 3 seats locked until 6 PM.

OUTPUT RANK

#1
Group A - Priority Pass solo
Quick turnover + flight departing soon

#2
Group B - Amex Platinum couple
Balanced value and capacity

#3
Group C - LoungeKey party of 3
Highest commercial value

Without PriModel: a group of 3 arriving 2 min earlier freezes the lounge until 6 PM. With PriModel: the solo creates immediate turnover. 2 admissions instead of 1.

Test Case 2: The Fairness Multiplier

GLOBAL STATE

4 Seats Available | Mode: Balanced

Group A - Priority Pass couple
Party: 2 | $27/pp | Total: $54
Flight: 5:15 PM (1h) | Wait: 35 min

Group B - Amex Platinum party of 4
Party: 4 | $25/pp | Total: $100
Flight: 6:15 PM (2h) | Wait: 5 min

Group C - LoungeKey solo
Party: 1 | $24/pp | Total: $24
Flight: 7:15 PM (3h) | Wait: 40 min

ANALYSIS

Why Fairness Dominates

Fairness scores:
Group A: 35 min wait = score 70, 4 skips = 55
Fairness composite: 64
Group B: 5 min wait = score 10, 0 skips
Fairness composite: 6

A 64 vs 6 gap (58 points) on Fairness.
$100 revenue cannot overcome passengers waiting 7x longer.

OUTPUT RANK

#1
Group A - Priority Pass couple
Long wait + flight departing soon

#2
Group C - LoungeKey solo
Longest waiting group

#3
Group B - Amex Platinum party of 4
Highest commercial value

Without PriModel: premium programs keep jumping the queue. With PriModel: equilibrium. The $100 group climbs as their wait increases, but cannot skip passengers waiting 7x longer.

Test Case 3: Short Visits Are Optimal

GLOBAL STATE

2 Seats Available | Mode: Balanced

Group A - LoungeKey couple
Party: 2 | $24/pp | Total: $48
Flight: 5:45 PM (1h 30min) | Wait: 20 min

Group B - Priority Pass couple
Party: 2 | $27/pp | Total: $54
Flight: 7:15 PM (3h) | Wait: 15 min

Group C - Amex Platinum solo
Party: 1 | $25/pp | Total: $25
Flight: 4:55 PM (40 min) | Wait: 10 min

ANALYSIS

No Taper. Short Visit = Optimal.

Group C at 40 min TTD: Urgency score = 90
(31-45 min range in lookup table)
No penalty. Passenger uses entitlement, lounge collects $25, seat frees in ~10 min.

Group A at 90 min: Urgency = 65
A wins #1: Fairness (20 min) outweighs C's urgency advantage in Balanced mode (W3=0.30 > W4=0.20).

OUTPUT RANK

#1
Group A - LoungeKey couple
Longest waiting + flight within window

#2
Group C - Amex Platinum solo
Low seat impact + near departure

#3
Group B - Priority Pass couple
Moderate commercial value

PriModel does not penalize short visits. A 10-minute lounge visit is a collected fee + immediate seat turnover. Staff sees departure badge and can override.

Test Case 4: Revenue Optimization When Capacity Allows

GLOBAL STATE

5 Seats Available | Mode: Revenue Focus (W1=0.45)

Group A - LoungeKey solo
Party: 1 | $24/pp | Total: $24
Flight: 6:15 PM (2h) | Wait: 20 min

Group C - Priority Pass couple
Party: 2 | $27/pp | Total: $54
Flight: 6:15 PM (2h) | Wait: 18 min

Group B - Day Pass party of 5
Party: 5 | $35/pp | Total: $175
Flight: 6:15 PM (2h) | Wait: 15 min

ANALYSIS

Why Revenue Dominates Here

All 3 groups: same TTD (2h).
Urgency = 45 for all. Neutralized.

Revenue scores:
B = 100 ($35/pp) | C = 80 ($27) | A = 60 ($24)

Revenue Focus (W1=0.45):
B: 45.0 | C: 36.0 | A: 27.0

The 18-point revenue gap (45 vs 27) overcomes capacity differences.

$175 captured in a single transaction.

OUTPUT RANK

#1
Group B - Day Pass party of 5
Highest commercial value

#2
Group C - Priority Pass couple
Balanced value and capacity

#3
Group A - LoungeKey solo
Longest waiting group

With equal departure times, Revenue becomes the sole differentiator. Revenue Focus captures $175 in a single admission. Group A still in top 3. If wait crosses 60 min, hard override fires.

Test Case 5: The Peak Throughput Machine

GLOBAL STATE

GRU Friday 4:15 PM | 95% capacity, 2 seats | Mode: Efficiency Focus

A: PP solo | $27 | LA 635 SCL 5:00 PM | Wait: 38 min
E: PP solo | $27 | AR 1161 EZE 5:10 PM | Wait: 30 min
B: Airline solo | $27 | AV 109 BOG 5:05 PM | Wait: 22 min

F: Airline solo | $27 | LA 2480 LIM 5:15 PM | Wait: 18 min
C: Day Pass couple | $70 | AA 920 MIA 7:15 PM | Wait: 42 min
D: LK party of 4 | $96 | AA 920 MIA 7:15 PM | Wait: 15 min

ANALYSIS

The Throughput Machine

Solos: 15-25 seat-min each. Urgency: 80-90.
Group C: 300 seat-min. Group D: 360 and does not fit.
Capacity gap: 12-20x.

4:15 call A. 4:20 call B. 4:32 call E. 4:42 call F.

4 admissions through 2 seats in 45 minutes.
$108 collected vs $70 from a single long-stay.

Group C protected: Max Wait override fires at 60 min.

OUTPUT RANK

#1

Group A - PP solo, SCL

Quick turnover + flight departing soon

#2

Group E - PP solo, EZE

Quick turnover + flight departing soon

#3

Group B - Airline solo, BOG

Quick turnover + flight departing soon

4 admissions through 2 seats in 45 min. $108 vs $70 from a single long-stay admission. The throughput machine that no manual FIFO process can replicate.

Algorithm

Part III

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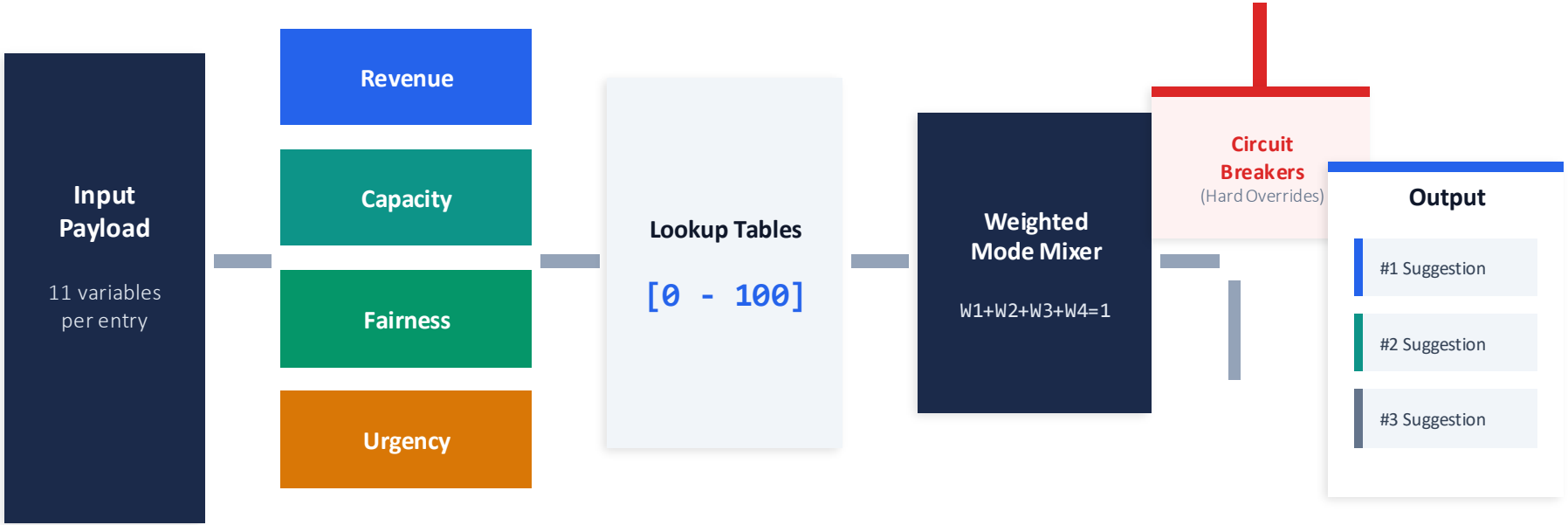
System Architecture: Processing Pipeline

Each waitlist entry is independently evaluated across four parallel dimension engines, normalized to [0, 100] via parametrized lookup tables, weighted, and filtered through hard overrides.



Architectural principle: Parallel independent evaluation. Each entry concurrently scored across 4 dimensions. Parametrized lookup tables produce stable scores on [0, 100]. Hard overrides bypass the weighted mixer when triggered.

5.1 Revenue Score (Lookup Table)

Value/pp Lookup Table

```
$35+ = 100  (Day Pass)
$27-34 = 80  (Priority Pass)
$25-26 = 70  (Amex Platinum)
$20-24 = 60  (LoungeKey)
$15-19 = 45 | $10-14 = 30 | $0-9 = 15
Unknown = 50  (neutral)
```

Revenue Data Waterfall

Tier 1	Deal-based visit cost (Phase 3)
Tier 2	Benefit card fee (Phase 2)
Tier 3	Lounge default value/pp (Phase 2)
Tier 4	Neutral: score = 50 (Phase 2)

Example from Scenario 4 (Revenue Focus)

Group	Program	Party	Value/pp	Total	RevenueScore
B	Day Pass	5	$35	$175	100
C	Priority Pass	2	$27	$54	80
A	LoungeKey	1	$24	$24	60

In Revenue Focus (W1=0.45), the gap between B (100) and A (60) contributes 0.45 x 40 = 18 points. Decisive.

5.2 Capacity Score (Lookup Table)

```
Party Size: 1=100  2=85  3=70  4=55  5=40  6+=25

Est Stay: 0-20m=100  21-40=85  41-60=70
61-90=55  91-120=40  121-150=25  151-180=15  180+=10

Composite:
CapacityScore = (Party x 0.4) + (Stay x 0.6)
```

Occupancy Pressure Modifier

Low (<50%)	x 0.8	Capacity matters less
Medium (50-80%)	x 1.0	Standard scoring
High (>80%)	x 1.2	Strong small/short preference

Example from Scenario 1

Group	Party Score	Stay Score	Composite
A (PP, solo)	100	100 (20 min)	100
B (Amex, couple)	85	55 (90 min)	67
C (LK, group of 3)	70	15 (150 min)	37

5.3 Fairness Score (Lookup Table)

```
Wait Time: 0-5m=10  6-10=20  11-15=30  16-20=40  21-25=50  26-30=60  31-35=70  36-40=80  41-50=90  51-60=95  60+=100

Skips: 0=0  1=15  2=30  3=45  4=55  5=65  6=75  7=82  8=90  9=95  10+=100

Composite: FairnessScore = (WaitTime_Score x 0.6) + (Skips_Score x 0.4)
```

Example from Scenario 2

Group	Wait Score	Skip Score	FairnessScore	Calculation
C (LK, solo)	80 (40 min)	65 (5 skips)	74	80x0.6 + 65x0.4
A (PP, couple)	70 (35 min)	55 (4 skips)	64	70x0.6 + 55x0.4
B (Amex, party 4)	10 (5 min)	0 (0 skips)	6	10x0.6 + 0x0.4

Fairness is the most robust dimension: wait time is always available (from join_timestamp). Skips tracked automatically. No external data dependency.

5.4 Urgency Score (No Taper)

Time to Departure Lookup (NO TAPER):

```
0-15 min = 100  (max urgency, fast turnover)
16-30 = 95  |  31-45 = 90  |  46-60 = 80
61-90 = 65  |  91-120 = 45  |  121-150 = 25
151-180 = 10  |  180+ = 0

Design: passenger uses entitlement, lounge collects fee, seat frees fast.
Everyone wins.
```

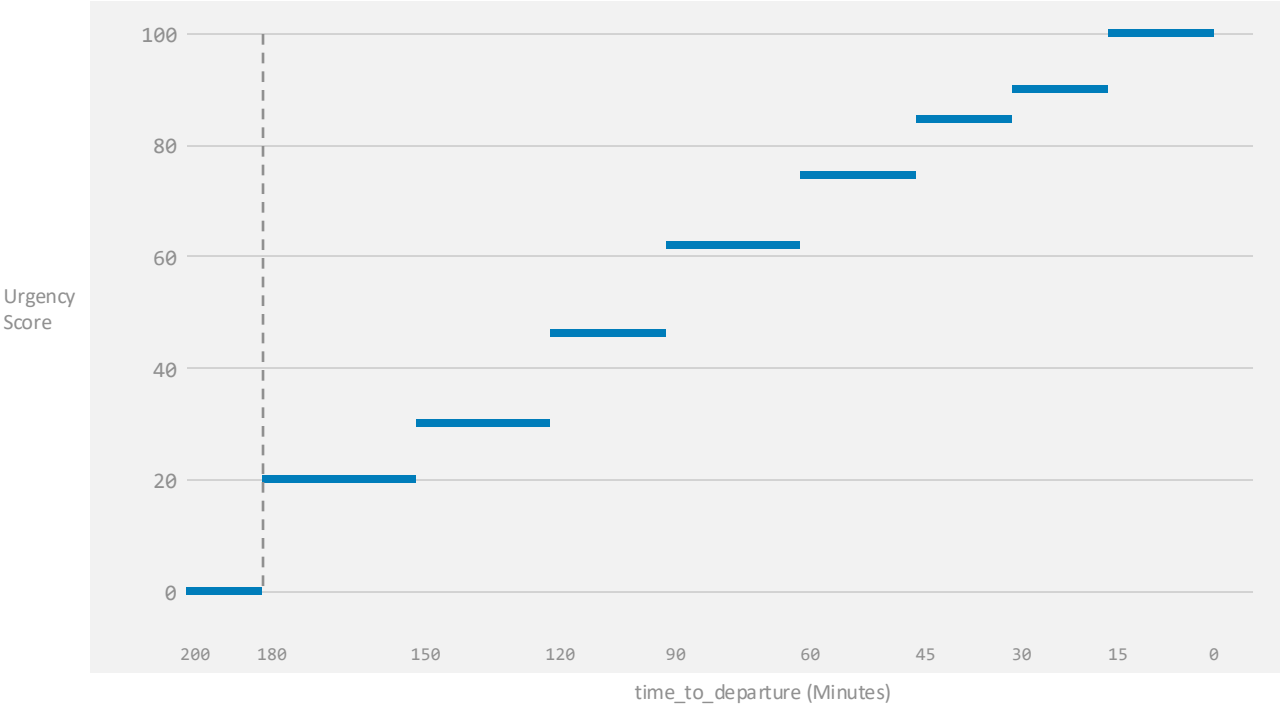
Flight Status Modifiers

On Time	Standard (x1.0)
Delayed	Recalculate TTD
Boarding	Force to 100
Departed	PriModelScore = 0
Not Found	Score = 50

Example from Scenario 5 (Peak Hour GRU)

Group	TTD	Range	UrgencyScore
A (PP) - SCL	45 min	31-45	90
B (Airline) - BOG	50 min	46-60	80
E (PP) - EZE	55 min	46-60	80
F (Airline) - LIM	60 min	46-60	80
D (LK) - MIA	180 min	180+	0

5.4 Urgency: No Taper. Short Visits Are Optimal.



Design Decision

Alternative considered:
Taper below 45 min (reduce score for very short visits).

Rejected: short visits collect the fee and free the seat. No reason to penalize. A 10-min visit is pure throughput.

Status Modifiers

Boarding: Force to 100
Delayed: Recalculate TTD
Departed: Score = 0
Not Found: Score = 50

6. Score Composition

```
PriModelScore = (W1 x Revenue) + (W2 x Capacity) + (W3 x Fairness) + (W4 x Urgency)
```

W1+W2+W3+W4 = 1.0 | All sub-scores [0, 100] | PriModelScore: 0-100 | Top 3 shown as suggestion cards

Why results are decisive, not marginal

Revenue
Day Pass vs LK
100 vs 60
40-pt gap

Capacity
Solo 20m vs Group 3h
100 vs 37
63-pt gap

Fairness
40 min vs 5 min
80 vs 10
70-pt gap

Urgency
TTD 45m vs 180+
90 vs 0
90-pt gap

Reasoning Labels (on suggestion cards)

Revenue dominant: "High commercial value"

Capacity dominant: "Quick turnover opportunity"

Fairness dominant: "Longest waiting group"

Urgency dominant: "Flight departing soon"

7. Hard Override Rules

These bypass the composite score. No amount of weight tuning changes the outcome.

P1 **Max Wait** `Trigger: wait_time > 60 min` **Effect: Score = 999**

No passenger waits more than 60 min without being prioritized.

P2 **Starvation Guard** `Trigger: consecutive_skips > 10` **Effect: Score = 998**

A passenger skipped 10 times must be surfaced regardless of value.

P3 **Flight Departed** `Trigger: flight_status = Departed` **Effect: Score = 0**

Excluded from suggestions. Flight has left.

P4 **Program Cap** `Trigger: Last 3 same access_program` **Effect: Score x 0.5**

Prevents one program from monopolizing admissions.

Program Cap edge cases: queue with only one program = cap suspended. Capped entry with Max Wait Override = override wins (999). Cap counts admissions only.

7. Circuit Breakers: Override Bypass Architecture

Deterministic rules that fire before the composite score. They bypass the weighted scoring pipeline entirely.

Override Bypass Layer

Trigger 1: Max Wait
>60 min
Score = 999

Trigger 2: Starvation
>10 skips
Score = 998

Trigger 3: Departed
status=Dep
Score = 0

Trigger 4: Prog Cap
3x same
x 0.5

PriModel Compiler: PriModelScore = Sum(W x Dimension)

Penalty Multiplier — **Final Score** — **Staff UI Top 3**

System Focus Modes: Weight Manipulation Matrix

	Efficiency Focus	Balanced Focus	Revenue Focus
Capacity (Navy)	Heavily dominates. Max throughput.	Standard scaling.	Standard scaling.
Fairness (Forest Green)	Moderate.	Anchor. Long waits always win ties.	Acts only as a tie-breaker.
Urgency (Amber)	Standard scaling.	Moderate.	Acts only as a tie-breaker.
Revenue (Crimson)	Acts only as a tie-breaker.	Standard scaling.	Primary driver. 45% of formula.

Hard constraint: W3 (Fairness) never below 0.10 in any mode or custom config.

8.1 Optimization Modes

Mode	W1 Revenue	W2 Capacity	W3 Fairness	W4 Urgency	When to Use
Balanced	0.25	0.25	0.30	0.20	General purpose. Safe default. Fairness slightly dominant.
Revenue Focus	0.45	0.25	0.15	0.15	Commercial yield KPIs. Revenue primary signal.
Efficiency Focus	0.15	0.40	0.20	0.25	Peak hours, long queues. Capacity dominant.
Fairness First	0.10	0.20	0.50	0.20	After complaints. Wait time overwhelming factor.

Hard constraint: W3 (Fairness) never below 0.10 in any mode or custom config.

8.2 Architecture Requirement: Full Parametrization

Every numerical value in this model must be configurable per lounge via admin UI, without code changes or deployments. Includes: dimension weights, thresholds, buffers, multipliers, and internal ratios. A weight change from 0.25 to 0.30 should take effect immediately after the admin saves.

The active mode is visible in the staff UI as a toggle badge. Admin configures; staff sees which mode is active but does not change it.

8.3 Configuration Parameters (1/2)

Score Weights

Parameter	Default	Description
optimization_mode	"balanced"	balanced | revenue_focus | efficiency_focus | fairness_first | custom
w1_revenue	0.25	0.00-1.00
w2_capacity	0.25	0.00-1.00
w3_fairness	0.30	0.10-1.00 (floor: 0.10)
w4_urgency	0.20	0.00-1.00

Fairness

Parameter	Default	Description
max_wait_hard_ceiling	60 min	Triggers Max Wait Override
max_consecutive_skips	10	Triggers Starvation Guard
fairness_wait_weight	0.6	Wait vs skip split

Urgency

Parameter	Default	Description
urgency_window_max	180 min	Above this, urgency = 0
(urgency_window_min)	REMOVED	No taper. Short visits optimal.

8.3 Configuration Parameters (2/2)

Capacity

Parameter	Default	Description
max_stay_limit	180 min	Max assumed lounge stay
boarding_buffer	30 min	Exit buffer before departure
default_stay_estimate	120 min	Fallback when flight data missing
capacity_pressure_mode	"auto"	auto (LMS) or manual (staff toggle)
occupancy_high_threshold	0.80	Above: high pressure (x1.2)

Revenue

Parameter	Default	Description
default_value_per_person	0	Flat fallback (Tier 3)
revenue_data_source	"auto"	auto | benefit_card | deal_pipeline

Program Cap

Parameter	Default	Description
program_cap_enabled	true	Enable/disable
program_cap_window	3	Consecutive same-program limit
program_cap_penalty	0.5	Score multiplier

UI

Parameter	Default	Description
suggestion_card_count	3	Cards shown to staff (1-5)
refresh_interval_seconds	60	Recalculation interval
show_debug_scores	false	Sub-scores on cards (testing)

Data Degradation

Part IV

COLLINSON | PRIORITY PASS™

9. How PriModel Handles Missing Data

Design principle: Missing data produces neutral scores (50), never penalties.

Always Available

Wait time (from join_timestamp)
Party size (mandatory enrollment)
Flight number (mandatory, resolves to TTD)
Consecutive skips (system-tracked)

Frequently Missing

Access program (passenger may not declare)
Value/pp (depends on access program)
Membership ID (Phase 3)

Impact by Dimension

Dimension	Fallback	Impact
Revenue	50 (neutral)	Other 3 dimensions decide. Most common at launch.
Capacity	120 min default stay	Minimal. Party size still differentiates.
Fairness	None needed	No impact. Always works. Most robust.
Urgency	50 if API fails	Rare. Flight number is mandatory.

9.5 Practical Implication at Launch

At launch, the model operates with strong Capacity, Fairness, and Urgency signals, but weak or absent Revenue.

Balanced

Effectively 3-dimension model (Cap + Fair + Urg) across 75% of weight, Revenue neutral.

Revenue Focus

Loses differentiating power. Behaves like Balanced until access program coverage improves.

Efficiency Focus

Largely unaffected. Revenue carries only 0.15 weight.

Fairness First

Largely unaffected. Revenue carries only 0.10 weight.

No cliff where the model fails. Only a gradient of increasing accuracy as revenue data coverage grows.

Cross-Industry References

Part V

COLLINSON | PRIORITY PASS™

10. Cross-Industry Validation

Multi-dimensional scoring for queue prioritization is well-established across industries.

Hospital ER (ESI)

Acuity + Resource consumption. Advisory. Fixed.

Theme Parks (Disney)

Revenue (paid tier) + Wait. Automated. Binary.

Airline Revenue Mgmt

Demand + TTD + Fare class. Dynamic bid prices.

Call Center Routing

Priority + Skills + Wait time. Routing rules.

Restaurant Queues

Wait time + Party size. Host decides. Basic rules.

Ride-Hailing (Uber)

ETA + Utilization + Revenue. Dynamic. Closest analogy.

10.7 PriModel in Context

Industry	System	Dimensions	Advisory?	Config Weights?
Hospital ER	ESI Triage	Acuity + Resources	Yes (nurse)	No (fixed)
Airlines	Revenue Mgmt	Demand + TTD + Fare	Partial	Yes (bid prices)
Restaurants	Waitwhile/Yelp	Wait + Party size	Yes (host)	Limited
Theme Parks	Disney Lightning	Revenue + Wait	No (auto)	No (binary)
Call Centers	Genesys/Webex	Priority + Skills + Wait	Partial	Yes
Ride-Hailing	Uber Dispatch	ETA + Rev + Coverage	Partial	Yes (dynamic)
Airport Lounges	PriModel	Rev + Cap + Fair + Urg	Yes (staff)	Yes (modes + custom)

PriModel combines best practices from each:

Multi-dimensional scoring of hospital triage, yield optimization of airline RM, party-size awareness of restaurant queuing, fairness balancing of theme parks, weighted routing of call centers, and context-dependent weight adjustment of ride-hailing dispatch.

No existing lounge management system offers this capability. This is a differentiation opportunity for Collinson's LMS.

Why This Benchmark?

1. **Not an invention from scratch.** Multi-dimensional scoring for queue prioritization is a well-established pattern.

2. **Validated by decades of use.** From emergency rooms (1998) to cloud computing (2009) to ride-hailing (2010).

3. **Billions in measured impact.** American Airlines: $1.4B over 3 years. AWS Spot: 90% cost savings. Disney: 40% per-capita spend increase.

4. **No lounge system offers this.** Waitwhile has basic rules. IEG America's AIMS has no queue management at all.

5. **PriModel is a differentiation opportunity** for Collinson's LMS, backed by cross-industry validation.

[Heterogeneous Demand] + [Fixed Capacity] ≠ [FIFO]

The Obsolete Standard

Chronological waitlists fail in capacity-constrained environments.

They waste perishable inventory, ignore customer value, and create bottlenecks when demand is heterogeneous.

The Algorithmic Standard

- Revenue
- Capacity
- Urgency
- Fairness

Market leaders rely on multi-dimensional prioritization.

From airlines to cloud computing, they route and sequence demand by simultaneously scoring interconnected variables.

This benchmark deconstructs the 10 algorithms that dictate modern capacity management.

The Four Pillars of Prioritization

THE INSIGHT

Across 40 years and 10 distinct industries, every successful prioritization model ultimately balances these four core dimensions.

REVENUE

Value capture, willingness to pay

CAPACITY

Resource fit, geometry, duration

Wait time, trust, starvation ceilings

FAIRNESS

Deadlines, time-to-departure

URGENCY

THE LESSON

Optimizing for just one dimension causes system failure.

Pure Revenue creates consumer backlash (Ticketing).

Pure Fairness wastes capacity (Restaurants).

Pure Urgency ignores resource constraints (Healthcare).

The secret is configurable, weighted balancing.

1

Hospital Emergency Triage (ESI)

Emergency Severity Index, developed 1998

Origin & History

- Developed in 1998 by Drs. Richard Wuerz and David Eitel at Brigham and Women's Hospital, Boston.
- Existing 3-level scales lacked granularity to distinguish between immediate intervention and high-resource patients.
- ESI Version 4 released in 2012; ENA acquired ownership in 2019, releasing Version 5 in 2023.
- Concept traces to 18th century: Baron Larrey, Napoleon's chief surgeon, first sorted wounded by severity, not rank.

How It Works

- 5-level algorithm with up to 4 decision points combining acuity (urgency) and resource consumption (capacity).
- Levels 1-2: binary overrides for life-threatening and high-risk situations.
- Levels 3-5: scored by expected number of resources consumed (2+, 1, or 0).
- Advisory-only: the nurse retains final decision authority.

Real-World Results

- 94% of U.S. emergency departments use ESI (2019).
- 139.8 million ED visits per year in the U.S. (2021).
- Inter-rater reliability kappa: 0.89 (strong agreement across nurses).
- Computerized ESI cut median triage time 53% (5.9 min to 2.8 min).

Analogy to PriModel

- Both combine urgency + resource/capacity dimensions (not urgency alone).
- Both are advisory: nurse/staff retains override power.
- Both use binary overrides for critical cases + multi-dimensional scoring for the rest.
- Key difference: ESI has fixed weights. PriModel offers configurable weights and optimization modes per lounge.

2

Airline Revenue Management

Born from American Airlines, 1982

Origin & History

- Post-deregulation (1978), low-cost carriers like People Express threatened full-service airlines.
- American Airlines' OR team (Thomas Cook, Barry Smith) framed seat allocation as optimization, 1982.
- DINAMO system implemented 1988. People Express collapsed 1987, attributed to AA's yield management.
- Built on SABRE, the reservation system co-developed by AA and IBM starting 1953 ($40M, 400 human-years).

How It Works

- Manages perishable inventory (empty seats on departure = zero revenue) across fare classes and time windows.
- Demand forecasting per segment (business, leisure, discount).
- Seat protection/nesting: protect seats for high-fare classes, release remainder for discounts.
- Overbooking based on predicted no-show rates. Time-to-departure dynamics adjust pricing in real time.

Real-World Results

- 1991 INFORMS Edelman Award. AA estimated $1.4 billion benefit over 3 years.
- Delta credited yield management for $300M/year additional revenue in the 1990s.
- National Car Rental saved from bankruptcy via adapted yield management.
- By 2000, virtually all major airlines, hotel chains, and car rental firms had adopted RM.

Analogy to PriModel

- Both optimize “who gets access” to capacity-constrained, perishable resources.
- Both use time-to-departure as a critical signal (closer = higher value/urgency).
- Both balance short-term revenue vs. long-term fairness for price-sensitive segments.
- Key difference: airline RM is automated (accepts/rejects bookings). PriModel is advisory (staff decides).

3

Hotel Revenue Management

Adapted from airlines, 1988

Origin & History

- Born from a conversation between AA CEO Robert Crandall and Marriott CEO Bill Marriott Jr.
- NYT (1988) documented hotels adopting yield management. AMR planned systems for Hilton, Marriott, Budget.
- Marriott was the first mover: daily demand forecasts and inventory recommendations for 160,000 rooms.
- Cornell's Sheryl Kimes published foundational research (1994, 2001) on consumer acceptance of dynamic pricing.

How It Works

- Core metric: RevPAR (Revenue per Available Room).
- Demand forecasting by date, room type, customer segment (business, leisure, group, contract).
- Rate fencing: different prices for different segments via restrictions (advance purchase, min stay, loyalty).
- Channel management (direct, OTAs, corporate, walk-in) based on commission rates and customer value.

Real-World Results

- Marriott: “Revenue management contributed millions to the bottom line.”
- Marriott global RevPAR grew 4%+ in 2024, all regions exceeding expectations.
- The field evolved from “how many rooms at each price” to “what price, which customer, which channel, when.”
- Exactly mirrors PriModel’s evolution from simple FIFO to multi-dimensional scoring.

Analogy to PriModel

- Same fundamental constraint: fixed capacity, perishable inventory, heterogeneous demand.
- Hotel RM balances occupancy vs. ADR. PriModel balances throughput vs. revenue priority.
- Both use “fencing” to segment demand (booking restrictions / access program identification).
- Key difference: hotel RM focuses on pricing. PriModel focuses on sequencing (who gets called next).

4

Restaurant Queue Management

Waitwhile, Yelp - 2010s

Origin & History

- Evolved from pen-and-paper waitlists to digital platforms in the early 2010s.
- Yelp built engineering-grade wait time prediction using XGBoost ML models (2019).
- Waitwhile, currently used by Airport Dimensions, provides configurable rule-based prioritization.
- However, Waitwhile has not invested in multi-dimensional scoring: rules remain basic, not weighted composites.

How It Works

- Party size: the primary capacity signal. A table for 2 and a table for 6 are fundamentally different resources.
- Wait time: the primary fairness signal. Guests expect roughly FIFO ordering.
- Table turnover: throughput signal. Quick parties free tables sooner.
- VIP prioritization (frequent visitors, high spenders) exists as a binary flag, not weighted score.

Real-World Results

- Yelp's XGBoost model significantly improved wait time prediction vs. host estimates.
- Waitwhile processes millions of waitlist entries annually (restaurants, healthcare, retail).
- The restaurant industry proved party size is the most impactful variable in queue optimization.
- Seating two 2-tops vs. one 6-top at a 4-top table: the core PriModel capacity dynamic.

Analogy to PriModel

- Both manage variable party sizes competing for fixed capacity with wait time fairness pressure.
- PriModel adds revenue (dollar value), urgency (flight deadline), and configurable optimization modes.
- Closest analogy: a high-end restaurant with VIP guests, large parties, and time-limited reservations.
- PriModel is what Waitwhile would be if it had flight data, revenue awareness, and optimization modes.

5

Theme Park Queue Optimization

Disney FastPass to Lightning Lane, 1999-present

Origin & History

- 1999: Disney created FastPass (free paper tickets with return time windows).
- 2013: $1 billion My Magic Plus investment digitized system into FastPass+ with MagicBands.
- 2020: COVID shutdown. FastPass+ did not return. Disney needed new revenue streams.
- 2021: Lightning Lane replaced FastPass as a paid service. The free queue bypass was gone.

How It Works

- Standby queue: traditional FIFO, available to all guests.
- Virtual queue (Boarding Groups): lottery for highest-demand attractions. Free but limited.
- Lightning Lane Multi Pass: $15-30+/day for shorter queues across multiple attractions.
- Lightning Lane Single Pass: a la carte, $7-25+ each for premium rides. Creates two-tier experience.

Real-World Results

- Disney parks: per-capita guest spending up 40% in Q1 2022 vs. Q1 2019. Lightning Lane was a key driver.
- Simulation: Genie+ increased overall park wait times by 5 min, gave payers 3.34 extra rides.
- Primary criticism: fairness. Non-paying guests experience systematically longer waits.
- Academic research modeled this as a revenue-fairness tradeoff using genetic algorithms.

Analogy to PriModel

- Both optimize throughput of fixed-capacity resources: attractions / lounge seats.
- Disney has binary paid/free. PriModel has configurable weighted scoring with convergence.
- PriModel's max wait override and starvation guard are the fairness safety nets Disney lacks.
- Key insight: monetizing queue priority works commercially but fails without structural fairness.

6

Call Center Skill-Based Routing

Genesys, Webex - 1990s to present

Origin & History

- ACD (Automatic Call Distribution) emerged in the 1970s with simple longest-waiting-agent routing.
- Skill-based routing evolved in the 1990s: not all agents and not all calls are equal.
- Spanish-speaking customer + billing issue needs an agent with both skills, not just the longest idle one.
- Genesys Cloud, Webex, Sprinklr now use multi-dimensional routing across interaction and agent dimensions.

How It Works

- Conversation Score: composite of arrival time + priority value for routing order.
- Priority Score: hierarchical 1 (highest) to 10 (lowest). Highest-priority interaction routed first.
- Agent proficiency: skill ratings 1-100 per category. Interactions matched by weighted skill requirements.
- Wait time: longest-waiting interaction is the tiebreaker when priority and skills are equal.

Real-World Results

- Genesys Cloud handles millions of interactions daily across thousands of contact centers.
- Skill-based routing: 10-25% improvement in first-call resolution.
- 15-30% reduction in average handle time vs. simple round-robin.
- Columbia University research validated that weighted multi-skill routing outperforms single-skill in efficiency and fairness.

Analogy to PriModel

- Call centers route interactions to agents; PriModel routes seats to groups. Direction inverted, architecture identical.
- Both use composite scores from multiple weighted dimensions with configurable priorities.
- Both have fairness mechanism (longest-waiting-first as tiebreaker).
- Key insight: multi-dimensional scoring outperforms simple FIFO in heterogeneous demand environments.

7

Ride-Hailing Dispatch

Uber - 2010 to present

Origin & History

- Uber launched 2010 with simple nearest-driver greedy algorithm. Worked at low volumes, not at scale.
- Evolved to batch matching: accumulates requests, solves assignment globally via bipartite graph.
- Michelangelo ML platform serves 10 million predictions per second at peak.
- Surge pricing acts as real-time market-clearing mechanism, equivalent to PriModel's optimization modes.

How It Works

- Geolocation + ETA: GPS, traffic, historical route data for actual pickup time prediction.
- Driver rating and history: destination-specific ratings, complaint history, compatibility scores.
- Vehicle type matching: riders matched to appropriate categories (UberX, Uber Black, etc.).
- Context-dependent weight adjustment: high demand prioritizes speed; off-peak optimizes coverage.

Real-World Results

- Millions of daily rides across 10,000+ cities worldwide.
- Batch optimization measurably reduced average wait times city-wide vs. greedy matching.
- Patent US20170011324A1 describes the weighted scoring system in detail.
- Michelangelo: 10M predictions/sec demonstrates industrial scale of multi-objective scoring.

Analogy to PriModel

- Closest analogy to PriModel. Both use weighted multi-objective scoring with context-dependent modes.
- Both balance revenue, efficiency, fairness (wait time), and operational constraints.
- Both advisory at final step: Uber suggests match, driver accepts/declines; PriModel suggests, staff overrides.
- Uber surge pricing = Revenue Focus mode. Off-peak coverage = Efficiency Focus mode.

8

Hotel Check-in Prioritization

Loyalty-tier operations, 2000s

Origin & History

- Revenue management (pricing) expanded into operational prioritization at the front desk.
- Hotel chains realized check-in prioritization directly affects satisfaction, loyalty retention, and throughput.
- Loyalty tier became the primary signal: Marriott Bonvoy Titanium, Hilton Diamond get priority processing.
- Essentially a queue prioritization system weighted by customer lifetime value.

How It Works

- Loyalty tier: primary signal. Higher-tier guests get priority processing, upgrades, flexible policies.
- Arrival time vs. room readiness: balancing “room ready” inventory against “guest waiting” fairness.
- Room assignment: matching preferences (high floor, quiet, king bed) to available inventory.
- Group vs. individual: conference groups create surge demand similar to lounge departure waves.

Real-World Results

- Marriott Bonvoy: nearly 228 million members worldwide.
- Loyalty programs are core competitive advantage precisely because they create tiered prioritization.
- Priority check-in for loyalty members drives higher retention and increased share of wallet.
- The system validates that commercial-value-based prioritization works in face-to-face hospitality.

Analogy to PriModel

- Same structural problem: multiple parties, limited resource, different commercial values, fairness expectations.
- Loyalty tier in hotels = access program tier in PriModel (proxy for revenue value).
- Group vs. individual capacity dynamics mirror lounge party size optimization.
- PriModel adds urgency (flight departure) and configurable modes, but the underlying logic is the same.

9

Cloud Computing Resource Allocation

AWS Spot Instances, 2009

Origin & History

- Concept traces to 1968 (Sutherland: “A Futures Market in Computer Time”).
- AWS launched EC2 in 2006. Had significant idle capacity that couldn’t be monetized without cannibalizing.
- 2009: AWS Spot Instances launched. Users bid on spare capacity. Resources granted while bid > spot price.
- 2017: AWS shifted from auction to provider-managed dynamic pricing with longer-term smoothing.

How It Works

- Price willingness: users set max price. System allocates to users above current spot price.
- Instance type flexibility: accepting multiple types/regions/zones improves allocation odds.
- Interruption tolerance: spot instances reclaimed with 2-min notice. Workloads must be fault-tolerant.
- Capacity pool diversification: balances allocation across types and zones for stability.

Real-World Results

- Spot Instances: up to 90% discount vs. on-demand pricing.
- Chegg achieved 70% reduction in EC2 costs via spot instances.
- Validated that dynamic allocation using multi-dimensional scoring (price, flexibility, tolerance) outperforms fixed-price FIFO.
- The spot market model inspired dynamic resource allocation across the tech industry.

Analogy to PriModel

- Cloud instances and lounge seats are both perishable capacity. Unused = zero revenue.
- Both use dynamic prioritization adjusting to supply-demand conditions.
- Both use graceful degradation: interrupted workloads migrate; PriModel falls back when data is missing.
- Key insight: market-aware, multi-dimensional allocation dramatically outperforms static FIFO.

10

Event Ticketing and Dynamic Pricing

Ticketmaster / Live Nation, 2010s-present

Origin & History

- Event ticketing transformed from fixed-price, first-come-first-served to demand-responsive pricing.
- Taylor Swift Eras Tour presale (Nov 2022): 3.5M registered, site crashed, 2.4M tickets sold in one day.
- Springsteen 2023: dynamic pricing pushed nosebleeds to $500+. Oasis dropped dynamic pricing after UK backlash.
- Taylor Swift herself rejected dynamic pricing for the Eras Tour, using a fixed-price model.

How It Works

- Verified Fan: registration + lottery system to filter bots, allocate access using engagement data.
- Dynamic pricing: real-time price adjustment based on demand per section.
- Tiered access: presale windows segmented by fan club, credit card holders, verified fans, general public.
- Seat location vs. price willingness: matching consumer WTP to available inventory across venue sections.

Real-World Results

- Eras Tour: $2B+ in ticket sales, largest-grossing tour ever. $5B estimated U.S. economic impact.
- Renaissance World Tour: $579M gross. Reportedly increased Sweden's inflation in May 2023.
- Live Nation 2023: $22.7B revenue (+36% YoY), 145M fans, 50,000+ events.
- Proved demand-responsive allocation works commercially but revealed fairness backlash when exploitative.

Analogy to PriModel

- Both allocate fixed-capacity resources with heterogeneous demand. Both face revenue vs. fairness tension.
- Ticketmaster dynamic pricing is like PriModel's Revenue Focus mode.
- But Ticketmaster lacks PriModel's fairness hard overrides. Fans can be priced out completely.
- PriModel's max wait override and starvation guard address the gap that generated Congressional hearings.

Comparative Overview

#	Industry	System	Dimensions	Advisory?	Configurable?	Origin
1	Hospital ER	ESI Triage	Acuity + Resources	Yes (nurse)	No (fixed)	1998
2	Airlines	Revenue Mgmt	Demand + TTD + Fare	Partially	Yes (bid prices)	1982
3	Hotels	Revenue Mgmt	Demand + Segment + Ch.	Partially	Yes (rate fences)	1988
4	Restaurants	Waitwhile, Yelp	Wait + Party size	Yes (host)	Limited	2010s
5	Theme Parks	Disney Lightning	Revenue + Wait	No (automated)	No (binary)	1999
6	Call Centers	Genesys, Webex	Priority + Skills + Wait	Partially	Yes (weights)	1990s
7	Ride-Hailing	Uber Dispatch	ETA + Util. + Revenue	Partially	Yes (dynamic)	2010
8	Hotels (Ops)	Check-in Priority	Loyalty + Arrival + Rev.	Partially	Limited	2000s
9	Cloud	AWS Spot	Price + Flex + Tolerance	No (automated)	Yes (bid/price)	2009
10	Ticketing	Ticketmaster	Demand + Price + Fan	No (automated)	Yes (dynamic)	2010s
	Lounges	**PriModel**	**Rev + Cap + Fair + Urg**	**Yes (staff)**	**Yes (modes)**	**2025**

What PriModel Takes From Each

Hospital ESI

Multi-dimensional scoring architecture, advisory design, binary overrides

Airline RM

Yield optimization of perishable inventory, time-to-departure signal, $1.4B proof

Hotel RM

RevPAR as core metric, segment-aware pricing, hospitality consumer acceptance

Restaurant Queuing

Party size as capacity variable, wait time as fairness signal

Disney

Fairness-revenue tension proof, consumers will pay for priority

Call Centers

Weighted composite scoring with configurable priorities outperforms FIFO

Uber

Context-dependent weight adjustment, bipartite matching, advisory final step

Hotel Check-in

Loyalty tier as CLV proxy, group vs. individual capacity dynamics

AWS Spot

Market-aware dynamic allocation, graceful degradation, 90% efficiency gains

Event Ticketing

Demand-responsive allocation power + lesson: embed fairness structurally

Configurable Optimization Modes

Revenue Focus

Revenue	95
Capacity	50
Fairness	55
Urgency	45

Throughput Focus

Revenue	45
Capacity	98
Fairness	50
Urgency	52

Equitable

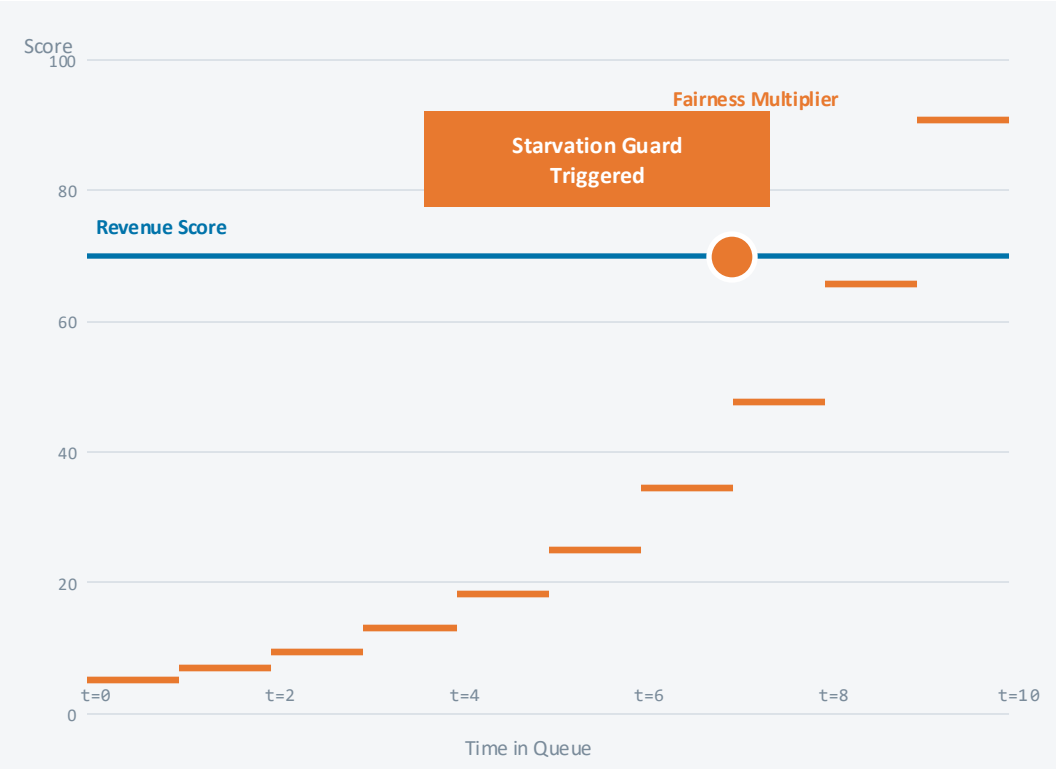
Revenue	40
Capacity	48
Fairness	99
Urgency	55

$$\text{Priority_Score} = (w1 \times R) + (w2 \times C) + (w3 \times F) + (w4 \times U)$$

Administrators select pre-configured modes based on time of day, flight wave, or lounge commercial contracts.

COLLINSON

The Fairness Constraint: Preventing Queue Starvation



The Disney/Ticketmaster Lesson

Consumers will pay for priority. But unrestrained revenue optimization destroys systemic fairness and user trust.

The Algorithmic Danger

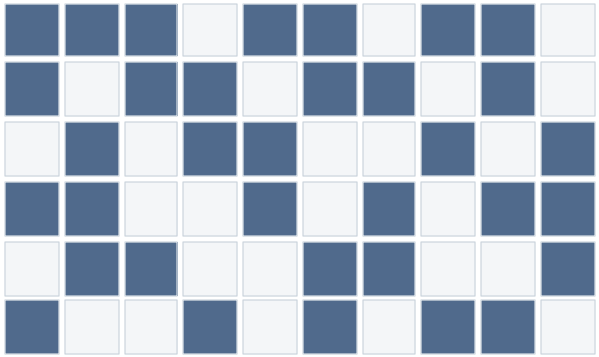
A purely revenue-weighted system permanently "starves" low-value users, leaving them queued indefinitely as high-value users bypass them.

PriModel Application

PriModel calculates a dynamic Fairness Multiplier. As wait time grows, fairness score scales exponentially, eventually forcing an override that breaks through the revenue ceiling.

The Geometry of Capacity

Waitwhile (Rules-based)



Party of 6 cannot fit

PriModel (Mathematical packing)

Parties of 2 fill gaps optimally

The Architectural Lesson

Yelp models prove party size is a primary mathematical variable. Simple rules-based waitlists fragment capacity. PriModel prevents fragmentation by mathematically prioritizing smaller groups when only scattered inventory remains.